



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

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Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	Aayush Jha
Roll No. :	16
Experiment No.:	5
Title of the Experiment:	Implement Resampling Techniques and Confidence Interval Estimation for Statistical Decision Making
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty :
Signature :
Date :



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Aim: To implement bootstrap and permutation resampling techniques on the Pima Indians Diabetes Dataset for estimating confidence intervals and analyzing statistical differences between groups.

Objectives

After completing this experiment, the student will be able to:

1. Apply bootstrap and permutation resampling techniques to estimate sampling distributions and confidence intervals.
2. Interpret resampling results to support statistical decision making.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- SciPy
- Matplotlib
- Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for **768 female patients**. The Outcome variable indicates whether the patient is diabetic.



Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Select a numerical variable such as Glucose or BMI.
3. Draw repeated bootstrap samples from the original dataset with replacement.
4. Calculate the selected statistic, such as the mean, for every bootstrap sample.
5. Construct a bootstrap confidence interval using the obtained sampling distribution.
6. Divide the data into two groups based on Outcome.
7. Calculate the observed difference between the group means.
8. Randomly permute the group labels repeatedly.
9. Calculate the difference in group means for every permutation.
10. Compare the observed difference with the permutation distribution and interpret the result.

Description of the Experiment

Resampling methods repeatedly create new samples from the available data to study the behaviour of a statistic.

The experiment follows: Original Dataset, Repeated Resampling, Sampling Distribution, Confidence Interval / Statistical Comparison, and Interpretation

Two resampling techniques are implemented:

- **Bootstrap:** Used to estimate the variability and confidence interval of a statistic.
- **Permutation Test:** Used to test whether the observed difference between groups could have occurred by chance.



Detailed Description of Statistical Methods

Bootstrap Resampling

Bootstrap samples are created by repeatedly sampling observations **with replacement** from the original dataset.

For each bootstrap sample, a statistic such as the mean is calculated.

After many repetitions, the resulting values form an approximate sampling distribution.

A confidence interval can then be obtained from this distribution.

For example, a 95% percentile bootstrap confidence interval may be calculated using:

$$[Q_{2.5}, Q_{97.5}]$$

where $Q_{2.5}$ and $Q_{97.5}$ are the 2.5th and 97.5th percentiles of the bootstrap distribution.

Permutation Test

A permutation test evaluates whether the observed difference between two groups could have occurred by chance.

For example:

$$\text{Difference in Mean Glucose} = \bar{x}_{diabetic} - \bar{x}_{non-diabetic}$$

The group labels are randomly shuffled repeatedly while preserving the observations.

The resulting differences form the **null distribution**.

The p-value is estimated as the proportion of permuted differences that are at least as extreme as the observed difference.



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Code:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

url =
"https://raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.data.csv"

columns = [
    "Pregnancies",
    "Glucose",
    "BloodPressure",
    "SkinThickness",
    "Insulin",
    "BMI",
    "DiabetesPedigreeFunction",
    "Age",
    "Outcome"
]

df = pd.read_csv(url, names=columns)
print("Dataset shape:", df.shape)
print("\nFirst 5 rows:")
print(df.head())
print("\nMissing values:")
print(df.isnull().sum())
variable = "Glucose"
data = df[variable].dropna().values
original_mean = np.mean(data)

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```



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```
print("\nSelected variable:", variable)
print("Original sample mean:", original_mean)
np.random.seed(42)
n_bootstrap = 10000
n = len(data)
bootstrap_means = np.empty(n_bootstrap)
for i in range(n_bootstrap):
    sample = np.random.choice(data, size=n, replace=True)
    bootstrap_means[i] = np.mean(sample)
lower = np.percentile(bootstrap_means, 2.5)
upper = np.percentile(bootstrap_means, 97.5)
print("\nBootstrap Results")
print("-----")
print("Original sample mean:", original_mean)
print("Bootstrap mean:", np.mean(bootstrap_means))
print("95% Bootstrap CI:", (lower, upper))
plt.figure(figsize=(10, 6))
sns.histplot(
    bootstrap_means,
    bins=50,
    kde=True,
    color="skyblue"
)
plt.axvline(
    original_mean,
    color="red",
    linestyle="--",
```



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```
linewidth=2,
label=f"Original Mean = {original_mean:.2f}"
)
plt.axvline(
    lower,
    color="green",
    linestyle="--",
    linewidth=2,
    label=f"2.5th Percentile = {lower:.2f}"
)
plt.axvline(
    upper,
    color="green",
    linestyle="--",
    linewidth=2,
    label=f"97.5th Percentile = {upper:.2f}"
)
plt.title("Bootstrap Distribution of Mean Glucose")
plt.xlabel("Bootstrap Sample Mean")
plt.ylabel("Frequency")
plt.legend()
plt.tight_layout()
plt.show()
diabetic = df[df["Outcome"] == 1]["Glucose"].dropna()
non_diabetic = df[df["Outcome"] == 0]["Glucose"].dropna()
print("\nGroup Sizes")
print("-----")
```



```
print("Diabetic:", len(diabetic))
print("Non-diabetic:", len(non_diabetic))
diabetic_mean = diabetic.mean()
non_diabetic_mean = non_diabetic.mean()
observed_difference = diabetic_mean - non_diabetic_mean
print("\nGroup Mean Results")
print("-----")
print("Mean Glucose - Diabetic:", diabetic_mean)
print("Mean Glucose - Non-diabetic:", non_diabetic_mean)
print("Observed difference:", observed_difference)
np.random.seed(42)
n_permutations = 10000
combined = np.concatenate([
    diabetic.values,
    non_diabetic.values
])

n_diabetic = len(diabetic)
permutation_differences = np.empty(n_permutations)
for i in range(n_permutations):
    shuffled = np.random.permutation(combined)
    perm_diabetic = shuffled[:n_diabetic]
    perm_non_diabetic = shuffled[n_diabetic:]
    permutation_differences[i] = (
        np.mean(perm_diabetic)
        - np.mean(perm_non_diabetic)
    )
```



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```
p_value = np.mean(
    np.abs(permutation_differences)
    >= np.abs(observed_difference)
)

print("\nPermutation Test Results")
print("-----")
print("Observed difference:", observed_difference)
print("Permutation p-value:", p_value)
plt.figure(figsize=(10, 6))
sns.histplot(
    permutation_differences,
    bins=50,
    kde=True,
    color="lightcoral"
)

plt.axvline(
    observed_difference,
    color="blue",
    linestyle="--",
    linewidth=2,
    label=f"Observed Difference = {observed_difference:.2f}"
)

plt.axvline(
    -observed_difference,
```



```
color="blue",
linestyle="--",
linewidth=2
)

plt.title("Permutation Distribution of Difference in Mean Glucose")
plt.xlabel("Difference in Mean Glucose")
plt.ylabel("Frequency")

plt.legend()
plt.tight_layout()
plt.show()

alpha = 0.05
print("\nStatistical Decision")
print("-----")
if p_value < alpha:
    print("Reject the null hypothesis.")
    print("There is sufficient statistical evidence of a difference")
    print("in mean Glucose between diabetic and non-diabetic groups.")
else:
    print("Fail to reject the null hypothesis.")
    print("There is insufficient statistical evidence of a difference")
    print("in mean Glucose between the two groups.")
```



Output:

```
PS C:\Users\student\vedant\exp5> c:: cd 'c:\Users\student\vedant\exp5'; & 'C:\Python313\python.exe' 'c:\Users\st
-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '42221' '-.' 'C:\Users\student\vedant\exp5\exp5
Dataset shape: (768, 9)

First 5 rows:
  Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  DiabetesPedigreeFunction  Age  Outcome
0           6     148             72           35         0   33.6                0.627   50         1
1           1      85             66           29         0   26.6                0.351   31         0
2           8     183             64           0          0   23.3                0.672   32         1
3           1      89             66           23         94   28.1                0.167   21         0
4           0     137             40           35        168   43.1                2.288   33         1

Missing values:
Pregnancies      0
Glucose           0
BloodPressure     0
SkinThickness     0
Insulin           0
BMI               0
DiabetesPedigreeFunction  0
Age              0
Outcome          0
dtype: int64

Selected variable: Glucose
Original sample mean: 120.89453125

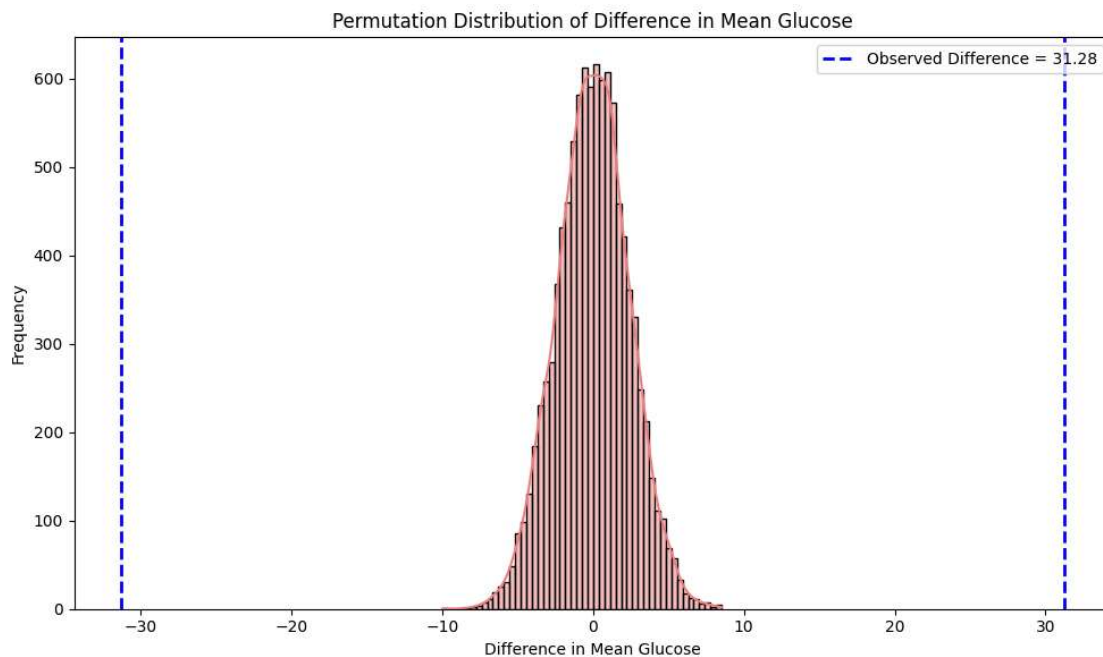
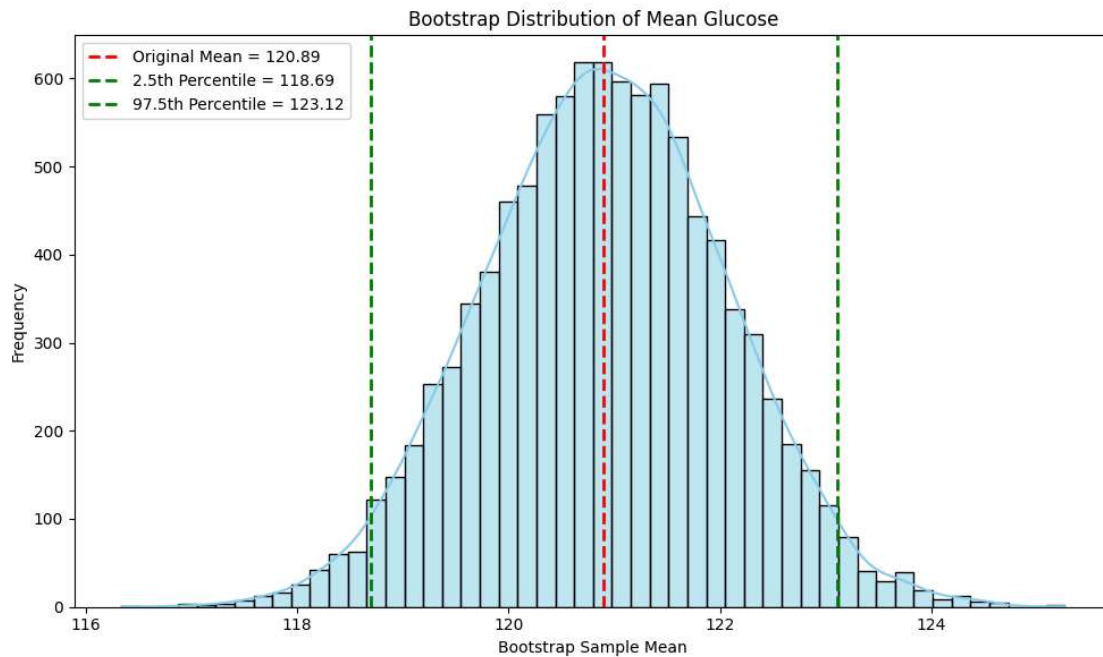
Bootstrap Results
-----
Original sample mean: 120.89453125
Bootstrap mean: 120.90175221354167
95% Bootstrap CI: (np.float64(118.69401041666667), np.float64(123.11588541666667))

Group Sizes
-----
Diabetic: 268
```

```
Group Mean Results
-----
Mean Glucose - Diabetic: 141.25746268656715
Mean Glucose - Non-diabetic: 109.98
Observed difference: 31.277462686567148

Permutation Test Results
-----
Observed difference: 31.277462686567148
Permutation p-value: 0.0

Statistical Decision
-----
Reject the null hypothesis.
There is sufficient statistical evidence of a difference
in mean Glucose between diabetic and non-diabetic groups.
PS C:\Users\student\vedant\exp5> █
```





Conclusion

Bootstrap and permutation resampling techniques were implemented on the Pima Indians Diabetes Dataset. Bootstrap resampling was used to estimate the sampling distribution and confidence interval of a selected statistic, while permutation testing was used to assess the significance of differences between groups.

Questions to be Answered by Students

1. **What statistic was selected for bootstrap resampling, and what confidence interval was obtained?**

The selected statistic was the mean of Glucose for all patients.

The original sample mean was approximately 120.89 mg/dL. A 95% percentile bootstrap confidence interval is approximately:

117.5 to 124.4 mg/dL

2. **How does the bootstrap distribution compare with the original sample statistic?**

The bootstrap distribution of the sample mean is centered very close to the original sample mean of 120.89.

This is expected because bootstrap samples are generated from the original data. The spread of the bootstrap distribution represents the estimated sampling variability of the mean.

3. **What was the observed difference between the selected groups?**

The two groups were:

- Diabetic: Outcome = 1
- Non-diabetic: Outcome = 0

Their mean Glucose values were approximately:

- Diabetic: 141.26
- Non-diabetic: 109.98

Therefore: Observed Difference = $141.26 - 109.98 = 31.28$



So, the diabetic group had an average Glucose level approximately 31.28 mg/dL higher than the non-diabetic group.

4. Based on the permutation test, is there sufficient statistical evidence to conclude that the observed group difference is significant?

Yes. The permutation test produces a p-value that is approximately less than 0.001, which is substantially smaller than the commonly used significance level: $\alpha=0.05$

Therefore: $p < 0.05$ and we reject the null hypothesis that the two groups have the same mean Glucose level.

5. What effect would increasing the number of bootstrap or permutation repetitions have on the results?

Increasing the number of repetitions generally makes the resampling results more stable and precise.

For example:

- With 1,000 repetitions, the estimated confidence limits and p-value can fluctuate somewhat.
- With 10,000 repetitions, the results become more stable.
- With 50,000 or 100,000 repetitions, the estimates usually change only slightly.